Alcohol Exposure and Cell State in the Served *Pseudomonas putida* Data

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KG 4.0 serves 32 *P. putida* datasets. Fifteen of them carry an alcohol growth or tolerance arm, or a perturbation of cell state, and Table 1 gives each one's genotype, environment, phenotype and genome as the served store holds them, with the environment axis broken out in Table 3. Every count in both tables is read off the store rather than off a loader docstring (Sec. 1).

Two findings come before the table. First, no served *P. putida* dataset measures isobutanol. A search of every loader module under `torchcell/datasets/pputida/` returns no match, and the two isobutanol datasets in the release are Lopez 2024, which is *S. cerevisiae* (Sec. 2). Second, the alcohol arms that do exist split into two kinds that a query must not merge: the transposon compendia feed an alcohol as the sole carbon source, which measures catabolism, while Lim 2022 and Lim 2025 apply one as a stressor (Sec. 3).

Four of the fifteen rows are transposon screens, and they are the largest by three orders of magnitude. Sec. 5 sets out what a transposon record means here, since the genotype is deliberately a gene-level aggregate rather than a clone and the phenotype deliberately avoids the fitness class.

Sec. 6 turns to the task these rows were surveyed for: predicting the final design-build-test-learn round of the Carruthers isoprenol campaign from its earlier rounds. It first lists all seventeen served datasets in which isoprenol appears, grouped by the role it plays, because only one of those roles is a label for this task and the largest block by far is another. It then tests whether any served gene-level readout tracks the campaign's per-target titer effect. None does reliably, including Yunus 2026, which shares 120 of the campaign's 135 targets. It closes on a blocker, which is that the cycle label is absent from every served Carruthers record. Sec. 7 closes with what the schema cannot yet hold.

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1 Where the numbers come from ✗

Two layers, and the split is what makes the tables auditable. Every count is probed from the served store. The row set and the last column of Table 1 are curation, held as data in the generating script so the judgment is versioned with the numbers it sits beside.

The stores read. The probe reads the dev build tree, `$DATA_ROOT/data/torchcell/<dataset>/processed/`, which is the tree a loader writes. It is not the knowledge-graph tree under `database/data/torchcell/`, which the build user owns and which is read-only here. The two were checked against the build manifest of release 2026.10.17-4295f7ca: every record count in Table 1 equals the manifest's `n_experiments` for that dataset, which is the evidence that the probed stores are the served ones.

Records are pickled, not JSON. An LMDB value is a pickle, so a probe reads it with `pickle.loads`. Reading one as UTF-8 fails on the pickle protocol byte, which is how this was found.

Constants are interned. An `Environment`, a medium and a compound are each stored once in a sibling `processed/interned/` LMDB, and a record carries `{"$ref": <sha256>, "name": <hint>}` in their place. A probe that does not splice these back reads every environment as an opaque reference. The script splices recursively before touching any field.

Distinct environments are counted by value. The interned table is the tempting route to the count and it is the wrong one: it holds every interned constant, environments and media and compounds alike, and nothing guarantees each entry is still referenced. So the probe reads every record, splices its environment and hashes the spliced object, which makes two references to the same condition one environment. That costs a full scan of 2.7 million records and it is exact. The result is cached in `experiments/database/results/pputida_cell_state_probe.json` so rebuilding the document does not rescan.

The genome is on the reference, not the experiment. Every row's assembly in Table 1 is read from `reference.genome_reference`. The experiment-level `genome_reference` is null in all fifteen datasets. That is by design rather than a defect: the genome never gates a record, so a record does not carry one.

All fifteen pin the same assembly, `GCA_000007565.2`, while the strain on that assembly varies: KT2440 for the screens, and the engineered backgrounds IY1449b, IPL400 and CJ522 for the production and tolerance work. An engineered strain's episomal cassettes are carried on the genotype as perturbations rather than folded into the assembly, which is why one accession covers every row.

What the probe cannot see. Whether a transposon pool was barcoded, and whether an alcohol was fed as a carbon source or applied as a stressor, are properties of the experiment that no stored field distinguishes. Both are carried as curation, the first in the italic line under the genotype column and the second in prose in Sec. 3.

2 There is no served *P. putida* isobutanol data ✗

This is the first thing to establish, because the release does contain isobutanol datasets and they are the wrong organism.

The loaders. A case-insensitive search for *isobutanol* across the fifteen loader modules under `torchcell/datasets/pputida/` returns no file.

The release. The two isobutanol datasets in KG 4.0, `IsobutanolScreenLopez2024` and `IsobutanolValidatedLopez2024`, resolve in the build manifest to `torchcell/datasets/scerevisiae/lopez2024.py`. They are *S. cerevisiae*.

The Fitness Browser deposit. The deposit's own database carries no isobutanol experiment for any organism, so the gap is upstream of our ingest rather than something a loader dropped.

One false positive to know about. A text search for *isobutyr* in the Borchert 2024 environment agents hits DL-3-aminoisobutyric acid, which is an amino acid fed as a nitrogen source. Matching on compound name rather than on structure is what produces that hit; the InChIKey does not.

The closest served alcohols are the nine alcohol and diol carbon sources the transposon compendia carry, 2-methyl-1-butanol among them, and the butanol and isopentanol arms of Lim 2022. None is isobutanol. A query that needs isobutanol in *P. putida* is a query for a dataset we have not ingested, not one for a filter over what is served.

3 The fifteen rows ✗

Table 1 is the row table and Table 2 its totals. Two distinctions decide whether a row answers a given question, and neither is a stored field, so both are stated here rather than read off the table.

Catabolism is not tolerance. In the transposon compendia the alcohol arms are carbon-source perturbations: the alcohol is the sole carbon source at 10 mM, and the fitness value says whether disrupting a gene impairs growth *on* that alcohol. In Lim 2022 and Lim 2025 the alcohol is a stressor added to a glucose medium, and the readout says whether the strain tolerates it. A query that filters on the compound alone returns both and they answer different questions.

Scale is not information. Almost every record in Table 2 is in the two pooled compendia, which that table reports as a share, because a pooled screen multiplies one condition by several thousand genes. The engineering rows at the bottom of the table hold single digits of records and carry the phenotype the strain-design work actually targets. The record count is a cost estimate, not a ranking.

The environment axis is the narrow one. Fifteen datasets, and the distinct environment counts in Table 3 are dominated by the two compendia's carbon and nitrogen source panels. Outside them the condition axis is one or two environments per dataset, so almost every row is a genotype series measured in a single condition.

Table 1: The served *P. putida* datasets bearing on alcohol exposure or cell state, ordered by record count. Every column but the last is read off the served store (Sec. 1). *Genotype* gives the perturbation types the records carry, with what the probe cannot see added in italic. *Env.* is the number of distinct `Environment` objects, counted by value over a full scan, and Table 3 breaks each one into its media, factors and agents. *Phenotype* is the stored class, then its measurement type and assay type. *Genome* is the assembly the reference record pins; the experiment-level `genome_reference` is null throughout, by design, since the genome never gates a record.

Dataset	Records	Genotype	Env.	Phenotype	Genome	Why it is here
RbTnseq Borchert2024	1,466,920	transposon insertion *gene-level aggregate, one gene per record*	155	**environment_response** log2_ratio / pooled_competitive_growth_barcode	KT2440 GCA_000007565.2	The RB-TnSeq compendium, and it carries Thompson 2020's alcohol panel. Nine alcohol and diol carbon sources, 2-methyl-1-butanol among them. These arms are growth ON the alcohol as the sole carbon source, so they measure catabolism, not tolerance to a solvent stress
FitnessBrowser Putida	1,118,052	transposon insertion *gene-level aggregate, one locus per record*	146	**environment_response** log2_ratio / pooled_competitive_growth_barcode	KT2440 GCA_000007565.2	The Fitness Browser deposit over the same alcohol panel, on 4,778 loci, and it is the one compendium that carries a moderated t statistic per record alongside the fitness value
FitnessContrast Borchert2023	64,853	transposon insertion	12	**t_statistic** rbtnseq_gene_fitness_two_sample_t_test / None	KT2440 GCA_000007565.2	The significance test of a lignin-related stressor arm against its reference arm, stored as the t statistic rather than as a fitness value
EnvMetalTnseq Royet2025	21,583	transposon insertion *non-barcoded mariner pool, gene-level aggregate*	4	**environment_response** log2_ratio / other	KT2440 GCA_000007565.2	Cobalt, copper, zinc and cadmium at sub-inhibitory dose over 12 generations. A metal stress rather than a solvent one, and the cleanest served example of a graded cell-state perturbation on a transposon pool
GradedGene Essentiality Royet2025	11,458	transposon insertion	2	**essentiality_call** None / None	KT2440 GCA_000007565.2	The unstressed essentiality baseline the metal contrasts above are read against, so the two rows are used together or not at all
RbTnseq Borchert2023	10,824	transposon insertion	32	**environment_response** log2_ratio / pooled_competitive_growth_barcode	KT2440 GCA_000007565.2	The loci Borchert 2023 scored that the compendium eliminated. Disjoint from the compendium row by construction, so it adds fitness values rather than duplicating them
CampaignProteome Carruthers2025	465	crispr interference, heterologous pathway	1	**protein_abundance** dia_nn_top3_percent_of_proteome_mean / None	IY1449b GCA_000007565.2	Cell state under CRISPRi knockdown across the six automated design-build-test-learn cycles, as a proteome panel. The paired feature table for the isoprenol titer campaign
PutidaPrecise321 Lim2022	321	released identifier	118	**expression_tpm** rnaseq_tpm / None	CJ522 GCA_000007565.2	The only served butanol exposure, in the compendium's Fuel project, beside an isopentanol arm, a three-point growth-phase series, a multistress panel of salt, imipenem and peroxide, an ionic-liquid adaptation series and two bioreactor runs. The broadest cell-state axis in the served set

Table 1, continued

Dataset	Records	Genotype	Env.	Phenotype	Genome	Why it is here
IsoprenolSelection Menasalvas2025	58	crispr interference, deletion strain, heterologous pathway	1	**environment__ response** categorical / biosensor__ readout	KT2440 GCA__ 000007565.2	Isoprenol production coupled to growth through a pyrF biosensor, so the readout is a categorical enrichment call rather than a titer
Proteome Menasalvas2025	6	deletion strain, heterologous pathway, locus variant call, site variant call	2	**protein__abundance** dia__protein__counts__sum__ mean / None	KT2440 GCA__ 000007565.2	Growth phase against production phase over 2,090 loci, with the two phases stored as distinct environments rather than as one condition
ProteomeFold ChangeLim2025	4	deletion strain, locus variant call, site variant call	2	**protein__fold__change** dia__nn__top3__log2__fold__ change__welch__two__ sample__t__test / None	IPL400 GCA__ 000007565.2	The proteome response to isoprenol stress, as a log2 fold change with its Welch test, which is the closest served analog of a solvent tolerance signature
IsoprenolTolerance Lim2025	3	deletion strain, locus variant call	2	**environment__ response** log2__ratio / liquid__od__ growth	KT2440 GCA__ 000007565.2	The only served alcohol tolerance growth phenotype: specific growth rate of a tolerized lineage against wild type at the same alcohol dose
ProteomeLim2025	3	deletion strain, locus variant call, site variant call	1	**protein__abundance** dia__nn__top3__log2__mean / None	IPL400 GCA__ 000007565.2	Absolute abundances on the same tolerized lineages as the row above, so the tolerance phenotype has a proteome beside it
MetaboliteGrowth Phase Menasalvas2025	3	deletion strain, heterologous pathway, locus variant call, site variant call	1	**metabolite__level** lc__ms__intracellular__ concentration__uM / None	KT2440 GCA__ 000007565.2	Intracellular pool sizes in growth phase, the metabolite half of the same phase contrast
Metabolite ProductionPhase Menasalvas2025	3	deletion strain, heterologous pathway, locus variant call, site variant call	1	**metabolite__level** lc__ms__intracellular__ concentration__uM / None	KT2440 GCA__ 000007565.2	The paired production-phase pools, which is the environment the titer campaigns actually sample

Table 2: Totals over the rows of Table 1, generated with them.

Quantity	Value
Datasets	15
Records	2,694,556
Records in the two pooled transposon compendia	2,584,972
Their share of all records	95.9%
Distinct environments, summed over datasets	480
Distinct phenotype classes	7
Distinct genotype perturbation types	7
Distinct genome assemblies	1

4 Reading the environment column ✗

The `Environment` object served in all fifteen datasets carries the same seven fields: `media`, `temperature`, `perturbations`, `aerobicity`, `duration_hours`, `duration_generations` and `dilution_rate_per_hour`. Each perturbation names a factor from a controlled vocabulary and an agent resolved to a compound identity with an InChIKey and a PubChem identifier, so an alcohol arm is queryable by structure. Table 3 breaks each row into those parts.

Two caveats on that column, and both limit what a query over it can claim.

Lim 2022's media are condition labels, not recipes. The compendium releases a condition name per sample and not the medium, the additions or the timing, so the media count for that row counts labels. A join on medium composition across that row and any other is not available.

A temperature can be a typed absence. Lim 2025 reports no incubation temperature for its tolerance assay, its evolution flasks or its proteome cultures, so the field carries a `ProvenanceGap` naming where it was looked for rather than a guessed 30 °C. Table 3 prints that absence as a value, since treating it as missing data is what turns a recorded gap into an invented number.

Table 3: The environment axis of each row in Table 1. *Env.* is distinct `Environment` objects and *Media* distinct media names. *Factors* counts the environment perturbations carrying each factor, summed over records, so a dataset of many records shows large counts for a few conditions. *Agents* is distinct compounds resolved to an InChIKey and a PubChem identifier, which is what makes an alcohol arm queryable by structure rather than by string. *Temperature* prints every value the store holds, a typed absence included.

Dataset	Env.	Media	Factors (perturbations carrying each)	Agents	Temperature
RbTnseqBorchert2024	155	4	carbon 974,792, nitrogen 492,128, dropout 89,908, small mol. 492,128	108	30 °C
FitnessBrowserPutida	146	3	carbon 621,140, nitrogen 496,912, dropout 90,782, small mol. 277,124	97	30 °C
FitnessContrast Borchert2023	12	1	carbon 64,853, small mol. 64,853	13	30 °C
EnvMetalTnseqRoyet2025	4	1	small mol. 21,583	4	28 °C
GradedGeneEssentiality Royet2025	2	2	none	0	28 °C
RbTnseqBorchert2023	32	1	carbon 10,824, small mol. 9,417	13	30 °C
CampaignProteome Carruthers2025	1	1	small mol. 465	1	24 °C
PutidaPrecise321Lim2022	118	115	carbon 15	3	not reported
IsoprenolSelection Menasalvas2025	1	1	pH 58, small mol. 116	2	not reported
ProteomeMenasalvas2025	2	1	pH 6, small mol. 6	1	not reported
ProteomeFoldChange Lim2025	2	1	small mol. 2	1	not reported
IsoprenolTolerance Lim2025	2	1	small mol. 3	1	not reported
ProteomeLim2025	1	1	small mol. 3	1	not reported
MetaboliteGrowthPhase Menasalvas2025	1	1	pH 3, small mol. 3	1	not reported
MetaboliteProduction PhaseMenasalvas2025	1	1	pH 3, small mol. 3	1	not reported

5 What a transposon record means here ✗

Four rows of Table 1 are transposon screens and they hold almost every record in it, so the representation is worth stating exactly.

The experiment. A randomly barcoded transposon library is a pool in which each cell carries one transposon inserted at a random genomic position, with a random DNA barcode in the insert. An insertion inside a gene disrupts it, so each cell is a loss-of-function mutant of one gene. The pool grows competitively in a condition, barcodes are sequenced at time zero and at the end, and a gene's fitness is the log2 ratio of its barcode abundance after growth over time zero, averaged across that gene's insertion strains and normalized so the typical gene sits at zero.

The genotype is a gene, not a clone. Each record carries one `TransposonInsertionPerturbation` with `insertion_position`, `insertion_strand` and `barcode` left null. Those nulls are the representation, not missing data: the stored value is the gene-level aggregate over every insertion mutant of that gene, which is the quantity the release reports. Royet 2025 is the case that makes this unavoidable, since its mariner library is not barcoded at all and a clone is unresolvable in principle.

The phenotype is not the fitness class. These records use `EnvironmentResponsePhenotype`, whose measurement type is `log2_ratio` and whose assay type is `pooled_competitive_growth_barcode`. `FitnessPhenotype` would be the obvious class and it is wrong here: `validate_fitness` clamps a non-positive value to 0.0, which would erase every fitness defect in the compendium, since a defect is exactly a negative log2 ratio. The signed ratio therefore goes on the environment-response class.

Reliability travels with the value. The Fitness Browser row carries a moderated t in `environment_response_t_statistic`, and the loader derives a standard error as $\sqrt{(f/t)^2 - 0.1^2}$, where f is the fitness value. That makes it the one compendium row where a per-record weight is available, which matters for any model that should down-weight a gene measured in a sparse pool.

Both compendia do measure isoprenol, as a substrate. The alcohol panel includes isoprenol itself as the sole carbon source at 10 mM, in MOPS minimal medium. Table 7 gives the record counts, which are the largest isoprenol block in the served data by three orders of magnitude. Each record is one gene's fitness when the pool has nothing but isoprenol to grow on, so the arm names the catabolism genes: a gene whose disruption impairs growth on isoprenol is a gene the host uses to consume it.

That makes the arm an engineering target in its own right and in the opposite direction from a titer campaign. A producing strain that also catabolizes its product loses yield to its own metabolism, and deleting a catabolism gene is how that is stopped, which is the design de Siqueira 2025's parent strain already embodies. The compendia are where the candidate genes are, genome-wide, without a new experiment.

For titer prediction they are features, not labels. The substrate arm does not make these rows training data for Sec. 6's task, and three differences are why. The readout is competitive growth, not product titer, and a gene that raises titer can be fitness-neutral or fitness-costly. The perturbation differs in kind: a transposon insertion is a full disruption while CRISPRi is partial repression of a gene that is still present, so the sign may agree but the magnitudes share no scale. And the chassis differs, since the libraries are wild-type KT2440 and carry neither the mevalonate pathway nor its flux burden.

What they can supply is a descriptor. A gene's fitness across the compendium's many carbon and nitrogen sources is a cheap, genome-wide measure of how essential or buffered that gene is, available for nearly every CRISPRi target without a new experiment. *Hypothesis, untested:* that descriptor improves last-round titer prediction over the proteome features alone. Nothing here measures it, and it should be checked against a proteome-only model before it is relied on.

6 Predicting the last Carruthers round ✗

The task these rows were surveyed for: predict the final design-build-test-learn round of the Carruthers isoprenol campaign from its earlier rounds, using whatever other isoprenol data is served. Carruthers is a collaborator, so the campaign's own data is available; the question is which single dataset from outside it helps.

6.1 What can help the campaign ✗

Table 4 is the answer in one page. It holds the thirteen served isoprenol datasets that are not Carruthers', ranked by how much of that campaign's target set each one reaches. Carruthers' own four stores are the thing joined to, so none of them is a row.

The ranking column is a join, not a size. Of the 135 CRISPRi targets the campaign knocks down, it counts how many each dataset also *perturbs* and how many it reports a value *for*, and the two are kept apart because they buy different things. A perturbed target gives a second measurement of that target's effect, which is what tests a model. A measured target gives that target a feature, which is what feeds one.

Three groups come out of it, and the order is not the order of record counts:

- **Yunus 2026 alone perturbs the target set**, 120 of 135, which made it the transfer candidate (Sec. 6.3). Measured, its titer does not rank those targets the way the campaign does (Sec. 6.4).
- **Three proteome datasets measure most of it**, 111, 111 and 100 targets, in conditions the campaign never ran. They are feature sources, not labels.
- **The two transposon compendia cover 86 targets** with a fitness value for growth on isoprenol, which is a per-target prior available without a new experiment, and separately name the catabolism genes (Sec. 5).

Everything below those reaches four targets or fewer. Those rows earn their place by design rather than by overlap, and the last column says which. Overlap is a precondition, not evidence of transfer: whether any of these values moves with the campaign's titer is tested in Sec. 6.4, and none does reliably.

Table 4: The served *P. putida* isoprenol data that is NOT Carruthers', ranked by how much of that campaign's target set it reaches. *Rec.* is the dataset's isoprenol records. *Pert.* and *Meas.* count, of the 135 CRISPRi targets the campaign knocks down, how many this dataset also perturbs and how many it reports a value for, such as a proteome keyed by locus tag. The two are kept apart because only the first gives a target a second measurement of its effect; the second gives it a feature. All four columns are read from a full scan. *What one row gives you* and *How it helps the campaign* are the pick list in the generating script.

#	Dataset	Rec.	What one row gives you	Pert.	Meas.	How it helps the campaign
1	IsoprenolTiter Yunus2026	125	isoprenol titer in ug/mL per CRISPRi strain	120	0	A second titer for 120 of the 135 targets, in another medium, temperature and chassis. Its ranking of those targets does not track the campaign's (Table~\{}ref{tab:pcscorr}), so it says how condition-bound a target effect is rather than scoring a model trained on the early cycles.
2	ProteomeLim2025	3	absolute Top3 abundance per protein in a tolerized lineage	1	111	Abundances for 111 of the 135 targets under isoprenol stress, so a target's protein level is known in a condition the campaign never ran.
3	ProteomeFoldChange Lim2025	2	log2 fold change per protein under isoprenol stress, with a Welch test	1	111	The same 111 targets as a stress contrast, which separates a target that responds to isoprenol from one that is merely abundant.
4	Proteome Menasalvas2025	6	summed DIA protein counts per protein, growth or production phase	4	100	Abundances for 100 of the 135 targets split by growth and production phase, which is the axis the campaign's single endpoint cannot resolve.
5	FitnessBrowserPutida	9,556	log2 fitness per locus on the same panel, with a moderated t	86	0	A growth-on-isoprenol fitness value for 86 of the 135 targets, with a moderated t so a weakly measured gene can be down-weighted. A per-target prior that needs no new experiment.
6	RbTnseqBorchert2024	9,464	log2 fitness per gene, growing on isoprenol as the sole carbon source	86	0	The same prior over a slightly different locus set, and the arm that names the catabolism genes: deleting one stops the chassis consuming the product the campaign is optimizing.
7	MetaboliteGrowth PhaseMenasalvas2025	3	intracellular concentration in uM per metabolite, growth phase	4	0	Four shared targets, and pathway intermediates rather than product, which is where a flux explanation of a knockdown can be checked against something other than titer.
8	MetaboliteProduction PhaseMenasalvas2025	3	the same pools in production phase, the paired environment	4	0	The same four targets in the phase the campaign actually samples, so it is the closer of the two to the campaign's own condition.
9	IsoprenolTiter Banerjee2024	20	isoprenol titer in ug/mL per (strain, condition family)	3	0	Three shared targets, and a set of deletions and pathway swaps the campaign never tried. Useful as an out-of-design check, not as training rows.
10	IsoprenolTiterDe Siqueira2025	10	isoprenol titer in ug/mL per tolerized isolate	2	0	Two shared targets on evolved isolates with resequenced genotypes, so it says what tolerization does to titer where the campaign only varies knockdowns.
11	IsoprenylAcetateTiter Kang2026	7	titer in ug/mL per strain, isoprenol on 7 records and the ester on 19	1	0	One shared target. Its value is the design rather than the overlap: the one served case where draining the isoprenol pool on purpose is the goal.
12	IsoprenolTolerance Lim2025	3	log2 growth rate against wild type at the same isoprenol dose	1	0	One shared target. Three records, so it is a sanity check on whether a high-titer strain is tolerating its own product, not a source of signal.
13	IsoprenolSelection Menasalvas2025	58	enriched or not, one biosensor call per CRISPRi target and round	0	0	No shared target, but a growth-coupled assay that scales past what a titer campaign can measure, so it is the route to a target list larger than 135.
	All 13 datasets	**19,260**				

6.2 What the served data holds on isoprenol ✗

Seventeen of the 32 served *P. putida* datasets carry isoprenol, over 2.59 million records of which 20,264 are isoprenol records. Table 7 is the short list: what each one would be used for, what one label is, the held-out unit a score would be read on, and the other datasets posing the same task. It bands by the role isoprenol plays, because the role decides the question the dataset can answer and only one of the four is a label for this task:

- **As the measured product**, in five titer campaigns. These are the only candidate labels. Kang 2026 is in this group on a technicality worth knowing: it is an isoprenyl acetate campaign built on the isoprenol chassis, and 7 of its 26 records carry isoprenol as the product while 19 carry the ester.
- **As the sole carbon source**, in both transposon compendia, which is the largest isoprenol block in the served data and measures catabolism rather than production (Sec. 5).
- **As a stressor**, in the three Lim 2025 datasets.
- **As the pathway the strain carries**, in the proteome, metabolite and selection datasets, which measure something else on a strain whose genotype names the cassette. These join to a campaign by strain rather than carrying a titer.

Of the four titer campaigns outside Carruthers, three are ruled out before any comparison of conditions. Banerjee 2024 holds 20 records across five condition families, and none of its strains is a CRISPRi knockdown, so there is no target set to share. de Siqueira 2025 holds 10 records on acetate-tolerized isolates, again not CRISPRi. Kang 2026's 7 isoprenol records are controls in an ester campaign. That leaves one.

6.3 The candidate: Yunus 2026 ✗

`IsoprenolTiterYunus2026` is the candidate on design, and Table 8 is why. Yunus 2026 (doi:10.1016/j.ymben.2025.11.007) selects CRISPRi knockdown targets by FluxRETAP against intuition, in the engineered isoprenol-producing strain IY1452, and Carruthers is a co-author. Three things line up and two do not. The measured test of whether it transfers is Sec. 6.4, and it does not.

The target sets nearly coincide. Every target in the Yunus campaign but one is also a Carruthers target (Table 8). That is what makes the dataset unusual: it is not a different experiment that happens to share a product, it is a second measurement of nearly the same target set.

The readout is the same quantity. Same product, same unit, same quantification method, same 48 hour culture. No unit conversion and no method bridge stands between the two campaigns. The replicate structure is the one readout axis that differs, and the difference is in what was released rather than in what was run. Carruthers reports a per-strain count, 2 to 6 biological replicates, and every record carries a sample standard deviation. Yunus releases one number per strain with the replicate count given only as a 3-to-6 range and the dispersion only as error bars, so the loader stores the conservative lower end of that range and records the missing dispersion as a typed gap. The uniform 3 in Table 8 is therefore a resolved lower bound, not a reported count, and a weighting scheme must not read it as one.

The modality is the same, the chassis designation is not. Both repress a native gene by CRISPRi in an isoprenol-producing strain. The served records pin different chassis, IY1449b for Carruthers and IY1452 for Yunus (Table 8), and the Yunus loader deliberately does not borrow the IY1449b genotype for IY1452, because the two are different designations and the paper states no genotype for its own. A target's effect therefore need not mean the same thing in both.

The paired features exist. `CrispriPanelProteomeYunus2026` carries an absolute Top3 proteome over 253 accessions on the same strains, which is the same feature modality as `CampaignProteomeCarruthers2025` in Table 1. A proteome-to-titer map can be fit on both.

The conditions differ, and this decides how it is used. Carruthers runs at 24 °C in M9-NREL with 20 g/L glucose; Yunus runs at 30 °C in plain M9 salts. The titer ranges differ accordingly, and the Yunus records carry no uncertainty, because only one number per strain was released and the loader records that as a typed gap rather than inventing a dispersion.

So Yunus is not extra training rows for absolute titer. The two uses it was a candidate for were a held-out condition, where a ranking learned on the earlier Carruthers cycles is scored on Yunus, and a reproducibility bound on the label. Both rest on the two campaigns ranking the shared targets alike, and they do not: rank agreement over the 120 shared targets is indistinguishable from none (Sec. 6.4). The reproducibility question has a different answer than expected. Within Carruthers' first round the label is highly reproducible, and between the two campaigns it is not, so the target effect is bound to the condition, the chassis or both, and the served data cannot say which.

The runner-up is `IsoprenolSelectionMenasalvas2025` in Table 1, the only other CRISPRi and isoprenol pairing. Its readout is a categorical biosensor enrichment call rather than a titer, so it is a weaker label and not the first choice.

6.4 Does any served readout track the campaign? ✗

Overlap says another dataset reaches a target. It does not say that dataset's value for the target moves with the campaign's titer, and Table 5 tests that directly.

The campaign's effect is a real signal. The per-target effect is the titer of the campaign's single-knockdown strains, 123 strains over 121 targets. Matched to the raw mirror's cycle column by target set, 119 of them are DBTL0 and the other four match no cycle or several, so the effect is a within-round estimate with no batch term. Replicate error accounts for under 2 percent of the spread between targets: the mean released standard error is 7.0 ug/mL against a spread of 73.7 ug/mL, a reliability of 0.985. A weak correlation below is therefore not caused by noise in the campaign.

No served readout tracks it. Every served gene-level dataset outside the campaign was read as gene vectors, one per experiment, sample or strain, and each vector was correlated with the effect over the shared targets. None clears the bar once the number of vectors a dataset offers is accounted for:

- **Yunus 2026**, the largest overlap, ranks the 120 shared targets in unrelated order, $\rho = -0.10$. Its top quartile shares 8 of 30 targets with the campaign's, against 7.5 by chance, and its bottom quartile shares 3. Its titers run about five times higher, a median of 858 against 163 ug/mL. Its own knockdown depth does not predict the campaign's effect either.
- **Growth on isoprenol** in the transposon compendia gives ρ between -0.06 and $+0.01$ over 77 targets (Table 6).
- **The best of hundreds** is what hundreds of unrelated vectors produce. The best of 234 and 310 compendium experiments reaches $\rho = +0.28$ with a best-of p near 0.8, and the best of 321 PRECISE321 samples reaches $+0.25$ at 0.09.
- **Lim 2025's stress proteome fold change** is the one best-of p below 0.05, $\rho = +0.27$ at 0.028 over four lineages. Against the all-strain fit it falls to $+0.18$, so it is a lead, not a result.

One weak lead among the priors. Targets whose transposon mutants show a strong defect in many experiments give higher first-round titer, $\rho = +0.27$ and $+0.24$ in the two compendia, and essential targets have a higher median titer than nonessential ones (Table 6). It does not stand as a finding. The priors were not chosen in advance, the two compendia share most of their experiments, and the correlation falls below $+0.10$ against the all-strain fit. Hypothesis (untested): knocking down a growth-limiting gene diverts carbon from biomass to product, which would make a fitness prior useful for ranking first-round targets.

What it leaves. For predicting the last round, the served data outside the campaign offers features with no measured relation to the label, and the campaign's own earlier rounds remain the training signal. The cycle label that split needs is the blocker of Sec. 6.5.

Table 5: Does another served dataset's gene-level value track the Carruthers per-target titer effect? The effect is the DBTL0 single-knockdown titer, 121 targets. Each dataset is a set of gene vectors, one per experiment, sample or strain. *Vec.* is the number of vectors tested out of the total, where a vector is tested when it shares at least 15 targets. *Shared* is the median shared targets. ρ and p are the Spearman correlation and its single-test p-value for the best vector. *Best-of p* compares that best ρ against the best under 2,000 shuffles of the target labels, which is the honest p when a dataset offers hundreds of vectors. *All* is the same vector against the additive fit over all 502 strains. A dash means no vector shares enough targets.

Dataset	One value is	Vec.	Shared	ρ	p	Best-of p	All
ProteomeFoldChangeLim2025	log2 fold change against the parent, one vector per lineage	4/4	97	+0.27	0.0066	0.028	+0.18
PutidaPrecise321Lim2022	RNA-seq TPM, one vector per sample	321/321	121	+0.25	0.0053	0.090	+0.20
FitnessContrastBorchert2023	t statistic of a two-arm fitness contrast, one vector per contrast	13/13	87	−0.25	0.021	0.196	−0.14
IsoprenolTiterYunus2026	isoprenol titer per CRISPRi strain	1/1	120	−0.10	0.28	0.279	−0.03
ProteomeLim2025	absolute Top3 abundance, one vector per lineage	3/3	97	+0.10	0.31	0.394	0.00
CrispriKnockdownYunus2026	the target's own protein against the non-targeting control	1/1	99	+0.05	0.62	0.623	−0.02
EnvMetalTnseqRoyet2025	Tn-seq log2 fold change under a metal, one vector per metal	4/4	109	+0.10	0.32	0.732	+0.01
FitnessBrowserPutida	log2 transposon fitness per gene, one vector per experiment	234/234	77	+0.27	0.016	0.755	+0.15
RbTnseqBorchert2024	log2 transposon fitness per gene, one vector per experiment	310/310	77	+0.28	0.015	0.802	+0.15
ProteomeBanerjee2025	log2 LFQ intensity, one vector per strain	6/6	80	+0.04	0.73	0.963	+0.05
ProteomeMenasalvas2025	summed DIA protein counts, one vector per strain and phase	6/6	88	−0.04	0.72	0.967	−0.02
ProteomeLog10PercentDe Siqueira2025	log10 percent of proteome, one vector per isolate	17/17	73	−0.04	0.75	0.995	−0.05
ProteomeDeSiqueira2025	Top3 peptide signal, one vector per isolate	17/17	73	−0.03	0.79	0.998	−0.07
ProteomePercentDe Siqueira2025	percent of proteome, one vector per isolate	17/17	73	−0.03	0.79	0.999	−0.07
RbTnseqBorchert2023	log2 transposon fitness per gene, one vector per experiment	0/42	10	–	–	–	–
CrispriPanelProteome Yunus2026	Top3 signal on a 207-protein panel, one vector per CRISPRi strain	0/125	11	–	–	–	–
CrispriDifferentialProteome Yunus2026	fold change against the control, one knockdown strain	0/1	8	–	–	–	–

Table 6: Single tests against the same effect: the compendia's two growth-on-isoprenol experiments, named because the best-of row in Table 5 hides them, and aggregate gene priors, each one vector. The priors were not chosen in advance and the two compendia share most of their experiments, so their p-values are not corrected and are not independent.

Test	Shared	ρ	p	All
RbTnseqBorchert2024: growth on isoprenol, set16IT008	77	−0.06	0.62	
RbTnseqBorchert2024: growth on isoprenol, set16IT009	77	+0.01	0.9	
FitnessBrowserPutida: growth on isoprenol, Putida:set16IT008	77	−0.06	0.62	
FitnessBrowserPutida: growth on isoprenol, Putida:set16IT009	77	+0.01	0.9	
RbTnseqBorchert2024: mean fitness over all 310 experiments	77	−0.20	0.077	−0.09
RbTnseqBorchert2024: share of experiments with |fitness| above 1	77	+0.27	0.018	+0.09
FitnessBrowserPutida: mean fitness over all 234 experiments	77	−0.19	0.094	−0.06
FitnessBrowserPutida: share of experiments with |fitness| above 1	77	+0.24	0.034	+0.07
PRECISE321: mean log2(TPM + 1) over 321 samples	121	+0.11	0.25	+0.12
Royet 2025 essentiality call: median DBTL0 titer in ug/mL by class, essential 199 (n 32), growth advantage 162 (n 13), growth defect 210 (n 7), nonessential 153 (n 69); Kruskal–Wallis			0.063	

6.5 The blocker: the cycle label is not in the graph ✗

The task needs a cycle label, to train on the earlier rounds and test on the last one. No served Carruthers record carries one. Table 9 searches every record in all four Carruthers stores for a DBTL token anywhere in its serialization and finds none, and `screen_id` is empty throughout.

The loader's record identity is one per construct and DBTL cycle, so the cycle is implied by the record set and was never written onto a record. From the graph alone the final round cannot be separated from the earlier ones, which means the task as stated cannot be set up from a query.

It is recoverable. The raw mirror's Dryad culture metadata carries a `DBTL_Cycle` column, and Table 10 gives its distribution. The join is on the construct name, so recovering the label is a preprocessing step rather than a re-ingest. It is still a loader gap worth filing: a campaign whose entire structure is the cycle should not lose the cycle on ingest.

Table 7: The served *P. putida* isoprenol data as a modeling short list, banded by the role isoprenol plays in each store. *Rec.*, *Iso.*, *Strains* and *Env* are read from a full scan: records in the store, records in which isoprenol plays that role, distinct genotypes among those records, and distinct stored environments among them. *Task*, *Label*, *Split*, *Same task* and *Why* are the pick list in the generating script. A compound is matched on its InChIKey, so a record spelling it 3-methyl-3-buten-1-ol is found and DL-3-aminoisobutyric acid is not; the pathway band is the exception, since a cassette names the product it makes rather than carrying its structure. The 15 served *P. putida* datasets in which isoprenol never appears are not listed. emphRec. and
emphIso. are summed per band and overall;
emphStrains and
emphEnv are not, since each counts distinct values inside one store and two stores can hold the same strain or the same condition.

#	Dataset	Task	Label	Rec.	Iso.	Strains	Env	Split	Same task	Why
Isoprenol is the measured product										
1	IsoprenolTiter Carruthers2025	regression, strain x cycle	isoprenol titer in ug/mL per (CRISPRi construct, DBTL cycle)	502	502	432	2	held-out cycle, the last of six	IsoprenolTiter Yunus2026	The target campaign. The only served isoprenol data with a repeated design-build-test-learn structure, and the only one carrying a per-record sample SD, so a model can be weighted by measurement precision. The cycle the split needs is not on the record (Table 9).
2	IsoprenolTiter Yunus2026	regression, strain	isoprenol titer in ug/mL per CRISPRi strain	125	125	121	1	held-out condition, against Carruthers	IsoprenolTiter Carruthers2025	All but one of its targets are Carruthers targets, with the same GC-FID readout, on a different chassis designation (IY1452 against IY1449b) and in a different medium and temperature. Measured against the DBTL0 single knockdowns, it ranks the shared targets in unrelated order (Table 5), so it does not test a transferred ranking.
3	IsoprenylAcetateTiter Kang2026	regression, strain	titer in ug/mL per strain, isoprenol on 7 records and the ester on 19	26	7	11	13	held-out strains	IsoprenolTiter Banerjee2024, IsoprenolTiterDe Siqueira2025	An ester campaign on the isoprenol chassis, so its isoprenol records are the upstream intermediate rather than the target. Useful as the one served case where draining the isoprenol pool is the intent.
4	IsoprenolTiter Banerjee2024	regression, strain x medium	isoprenol titer in ug/mL per (strain, condition family)	20	20	12	9	held-out strains	IsoprenolTiterDe Siqueira2025, IsoprenylAcetateTiterKang2026	Model-guided deletions and pathway swaps rather than CRISPRi, so it shares no target set with the campaigns above. Every number is read from the article prose against the pinned OCR, since no measurement table was released.
5	IsoprenolTiterDe Siqueira2025	regression, strain	isoprenol titer in ug/mL per tolerized isolate	10	10	5	2	held-out strains	IsoprenolTiter Banerjee2024	Acetate-tolerized isolates of a catabolism-deficient parent, with a resequenced genotype per isolate. The one served row where the strain difference is an evolved variant call rather than a designed edit.
	Product subtotal			*683*	*664*					
Isoprenol is the carbon source: catabolism, not production										
6	RbTnseq Borchert2024	regression, gene	log2 fitness per gene, growing on isoprenol as the sole carbon source	1,466,920	9,464	4,732	1	held-out genes	FitnessBrowserPutida	Genome-wide loss-of-function fitness on isoprenol at 10 mM. It names the catabolism genes, which is the deletion target for stopping a producer consuming its own product, and it is the largest isoprenol block in the served data by three orders of magnitude.
7	FitnessBrowserPutida	regression, gene	log2 fitness per locus on the same panel, with a moderated t	1,118,052	9,556	4,778	2	held-out genes	RbTnseqBorchert2024	The same arm in the Fitness Browser deposit over a slightly different locus set, and the only one of the two carrying a per-record t statistic, so a gene measured in a sparse pool can be down-weighted.
	Carbon-source subtotal			*2,584,972*	*19,020*					
Isoprenol is the stressor										
8	IsoprenolTolerance Lim2025	regression, strain	log2 growth rate against wild type at the same isoprenol dose	3	3	3	2	held-out strains	ProteomeLim2025, ProteomeFoldChange Lim2025	The only served alcohol tolerance growth phenotype. Three records, so it is a check on a tolerance model rather than training data for one.
9	ProteomeLim2025	regression, protein x strain	absolute Top3 abundance per protein in a tolerized lineage	3	3	3	1	held-out strains	ProteomeFoldChange Lim2025, Proteome Menasalvas2025	The feature side of the tolerance phenotype above, on the same lineages, so the pair supports a proteome-to-tolerance map at three points.

Table 7, continued

#	Dataset	Task	Label	Rec.	Iso.	Strains	Env	Split	Same task	Why
10	ProteomeFoldChange Lim2025	regression, protein x strain	log2 fold change per protein under isoprenol stress, with a Welch test	4	2	2	1	held-out strains	ProteomeLim2025	The stress response as a contrast rather than a level, which is the closest served analog of a solvent tolerance signature.
	Stressor subtotal			*10*	*8*					
Isoprenol is the pathway the strain carries, and something else is measured										
11	CampaignProteome Carruthers2025	regression, protein x strain	DIA-NN Top3 percent of proteome per protein and campaign strain	465	465	401	1	held-out cycle	CrispriPanelProteome Yunus2026, Proteome Carruthers2025	The features paired with the target campaign, cycle by cycle. The natural input to a titer model, and the row whose cycle label is missing for the same reason the titer's is.
12	IsoprenolSelection Menasalvas2025	binary classification	enriched or not, one biosensor call per CRISPRi target and round	58	58	58	1	held-out targets	IsoprenolTiter Carruthers2025	Growth-coupled selection, so it scales where a titer assay does not. The loader is explicit that the titer it proxies is not measured per clone, so it is a ranking label and never a quantitative one.
13	Proteome Carruthers2025	regression, protein x strain	absolute abundance per protein and campaign strain	21	21	17	2	held-out strains	CampaignProteome Carruthers2025	The absolute-scale companion to the campaign proteome, on 21 records. A calibration check rather than a feature table.
14	ProteomeFoldChange Carruthers2025	regression, protein x strain	log2 fold change per protein against the per-cycle control	16	16	16	1	held-out strains	ProteomeFoldChange Lim2025	The campaign proteome as a contrast against the controls the cycles carry, which is the form a knockdown effect is read in.
15	Proteome Menasalvas2025	regression, protein x phase	summed DIA protein counts per protein, growth or production phase	6	6	3	2	held-out phase	Proteome Carruthers2025	The phase contrast over 2,090 loci, with the two phases stored as distinct environments rather than one condition, so the phase is queryable instead of implied.
16	MetaboliteGrowth PhaseMenasalvas2025	regression, metabolite x phase	intracellular concentration in uM per metabolite, growth phase	3	3	3	1	held-out phase	MetaboliteProduction PhaseMenasalvas2025	Pathway intermediates rather than the product, which is where a flux model is checked against something other than the titer.
17	MetaboliteProduction PhaseMenasalvas2025	regression, metabolite x phase	the same pools in production phase, the paired environment	3	3	3	1	held-out phase	MetaboliteGrowth PhaseMenasalvas2025	The phase the titer campaigns actually sample, so it is the arm that says what the cell looks like while it is making product.
	Pathway subtotal			*572*	*572*					
	All 17 datasets			**2,586,237**	**20,264**					

Table 8: The two served isoprenol titer campaigns, axis by axis, every value read off the store. The product is the same compound in both, isoprenol at InChIKey CPJRRXSHAYUTGL-UHFFFAOYSA-N and PubChem 12988. *Targets shared* is the intersection of the two CRISPRi target sets, counting the CRISPRi half of the genotype only, since a Carruthers record also carries the heterologous pathway.

Axis	Carruthers 2025	Yunus 2026
Records	502	125
CRISPRi targets	135	121
Targets shared by both campaigns	120	
Genotype perturbation types	crispr interference, deletion strain, gene addition, heterologous pathway	crispr interference
Chassis strain	IY1449b	IY1452
Medium	M9-NREL + 20 g/L glucose (10 mM ammonium sulfate + Teknova T1001; Carruthers 2025)	M9 salts (6.78 g/L Na2HPO4, 3 g/L KH2PO4, 1 g/L NH4Cl, 0.5 g/L NaCl)
Temperature	24 °C	30 °C
Duration	48 h	48 h
Replicates	2 biological replicates, 3 biological replicates, 6 biological replicates	3 biological replicates
Quantification	GC-FID	GC-FID
Titer unit	ug/mL	ug/mL
Titer range	0.00 to 771.2	1.59 to 1,495.0
Records carrying an uncertainty	502 of 502	0 of 125

Targets in Yunus 2026 and not in Carruthers 2025: PP_4129.

Table 9: The DBTL cycle label in the served Carruthers stores. *Naming a cycle* counts records whose full serialization matches `DBTL\s*\d` anywhere, so it is as permissive a search as the store allows.

Store	Records	Naming a cycle	With a screen id
campaign_proteome_carruthers2025	465	0	0
isoprenol_titer_carruthers2025	502	0	0
proteome_carruthers2025	21	0	0
proteome_fold_change_carruthers2025	16	0	0

Table 10: The same label in the raw mirror, which is the recovery path. Rows are cultures and include each cycle's controls, so these are not record counts; they establish that the label exists upstream and joins on `Line_name`.

Cycle	Culture rows
DBTL0	393
DBTL1	183
DBTL2	189
DBTL3	174
DBTL4	183
DBTL5	186
DBTL6	189
Total	**1,497**

7 What is released and not served ✗

One group of samples bearing directly on cell state is in no served dataset, and it is worth recording because a query cannot discover its absence.

The dropped reactor arms. Borchert 2024 releases 332 RB-TnSeq samples and the loader stores 310. The 22 it drops are the bioreactor and adaptation samples, under the rule `reactor_process_not_representable`: they differ by dissolved-oxygen set point, batch or fed-batch feed rate, sampling time, solvent overlay and adaptation day, and the loader's position is that collapsing distinct conditions into one environment is worse than not serving them. The arithmetic confirms which samples are served, since $310 \times 4{,}732$ is exactly the record count Table 1 reports for that row.

The residue is narrower than that rule implies. `Environment` already carries dissolved oxygen as a physical factor, plus `duration_hours`, `duration_generations` and `dilution_rate_per_hour`, so the oxygen set point, the sampling time and the adaptation day are representable on fields that exist today. Three things remain:

- **The solvent overlay has no representation.** A search of `torchcell/datamodels/schema.py` for overlay, two-phase, biphasic, dodecane and oleyl alcohol returns nothing. This affects 3 of the 22 samples, and it is the axis closest to in-situ product recovery for an alcohol, so it is the one worth a schema discussion.
- **Fed-batch feed rate is only partly covered.** `dilution_rate_per_hour` is documented as the controlled variable of a continuous chemostat, so reading a discontinuous fed-batch feed into it is a judgment call rather than a given.
- **The other 19 samples look admissible on existing fields.** That is a reading of the schema against the release metadata and not a verified admission. It needs owner sign-off and a rerun of the admission check before anyone acts on it.

What the Fitness Browser row leaves out. Of the deposit's 314 *P. putida* experiments the loader stores 234. It drops the 34 with an empty `pubId`, which includes the bioreactor series, and all 46 `VillalobosEscobedo23` samples, because every one is on R2A and the media library has no entry for it. The second exclusion is worth knowing when reading that row's agent count: the stress panel and methanol arms that deposit carries are not in the served set.